

Assignment - 3

Course Name: Algebraic Geometry

Indian Statistical Institute, Kolkata

Note: Throughout, k denotes an algebraically closed field unless otherwise specified, and a *variety* over k (or a k -variety) is any affine, quasi-affine, projective, or quasi-projective variety.

(1) Show that on any variety, there is a base for the topology consisting of affine open subsets.

(2) Let $\varphi : X \rightarrow Y$ be a morphism of varieties.

(a) Show that for every $p \in X$, φ induces a (local) homomorphism of local rings

$$\varphi_p^* : \mathcal{O}_{Y, \varphi(p)} \longrightarrow \mathcal{O}_{X, p}.$$

(b) If φ is an isomorphism, show that φ_p^* is an isomorphism of local rings for all $p \in X$. Is the converse true?

(c) If φ is dominant, show that φ_p^* is injective for each $p \in X$.

(3) Let $\varphi \in \text{End}_k(\mathbb{A}_k^n)$ be given by n polynomials $f_1, \dots, f_n \in k[x_1, \dots, x_n]$ Let

$$J = \det \left[\left(\frac{\partial f_i}{\partial x_j} \right)_{1 \leq i, j \leq n} \right]$$

be the *Jacobian* determinant of φ . Clearly $J \in k[x_1, \dots, x_n]$, known as the *Jacobian polynomial*. Show that if $\varphi \in \text{Aut}_k(\mathbb{A}_k^n)$, then $J \in k^*$, i.e, a non-zero constant polynomial¹.

(4) **Reading Assignment** (cf. Atiyah–Macdonald, *Introduction to Commutative Algebra*, Page 121; H. Matsumura, *Commutative Algebra*, Page 78; *The Stacks Project*, Lemma 10.110.3).

If $(A, \mathfrak{m}, \kappa)$ is a Noetherian local ring with maximal ideal $\mathfrak{m}$ and residue field κ , then

$$\dim_{\kappa}(\mathfrak{m}/\mathfrak{m}^2) \geq \dim A.$$

(5) Determine the dimension of the affine variety $X \subset \mathbb{A}_k^4$ given by the equations

$$xy - z^2 = x^2w^3 - y^6 = 0,$$

and find its singular points. Calculate the dimension of Zariski tangent spaces at the singular points.

Same for the variety given by

$$x^2w^2 - y^2z^3 = x^2yw - z^4 = y^3 - wz = 0.$$

(6) For every degree $d > 0$, and every $p = 0$ or a prime number, give the equation of a non-singular curve of degree d in $\mathbb{P}_k^2$ over a field k of characteristic p .

(7) Let X be a k -variety and $p \in X$. Prove that $\dim_k T_p(X) \geq \dim X$, with equality if and only if p is a non-singular point.

(8) If φ is the vertical projection of the parabola $C := \{y^2 - x = 0\} \subset \mathbb{A}_k^2$ onto the x -axis, show that the induced k -linear map

$$T_O(\varphi) : T_O(C) \rightarrow T_O(\mathbb{A}_k^1)$$

of Zariski tangent spaces at $O = (0, 0)$ is the zero map.

¹The converse is an unsolved problem, even for $n = 2$, famously known as the *Jacobian Conjecture*.

- (9) Let $X \subset \mathbb{P}_k^n$ be a projective variety. Let $f_1, \dots, f_t \in S = k[x_0, \dots, x_n]$ be homogeneous polynomials which define the ideal of X . Let $P \in Y$ be a point with homogeneous coordinates $P = (a_0 : \dots : a_n)$. Show that P is nonsingular on X if and only if the rank of the matrix

$$\left(\frac{\partial f_i}{\partial x_j}(a_0, \dots, a_n) \right)_{1 \leq i \leq t, 0 \leq j \leq n}$$

is $n - \dim(X)$.

- (10) Let $F \in k[x_0, x_1, x_2]$ be a homogeneous polynomial, and let $Y = Z(F) \subset \mathbb{P}_k^2$ be the corresponding algebraic set. Show that if F is reducible, then Y must be singular.